

Mixed-Integer Linear Programming Formulation for Rolling Stock Scheduling and Circulation

Quantum Annealing-Based Rolling Stock Scheduling Research Suite

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Abstract

This document presents the complete Mixed-Integer Linear Programming (MILP) formulation for the Rolling Stock Scheduling and Circulation Problem. The model simultaneously determines trainset-to-trip assignments, multi-unit consist coupling configurations, dead-head repositioning arcs, and periodic maintenance mileage bounds under rigid temporal and spatial station turnarounds.

1 Sets and Indices

- U : Set of rolling stock trainset units, indexed by $u \in U$.
- T : Set of revenue passenger timetable trips, indexed by $i, j \in T$, sorted by scheduled departure time.
- S : Set of stations across the railway network, indexed by $s \in S$.
- $D \subseteq S$: Subset of railway stations designated as overnight stabling depots.
- $D_{\text{maint}} \subseteq D$: Subset of depots equipped with specialized maintenance workshops.

2 Problem Parameters

2.1 Trip Attributes ($\forall i \in T$)

- $o_i, d_i \in S$: Origin and destination stations of trip i .
- $\text{dep}_i, \text{arr}_i \in \mathbb{R}_+$: Scheduled departure and arrival timestamps (minutes from midnight).
- $\tau_i \in \mathbb{R}_+$: Minimum required turnaround and cleaning buffer at destination d_i .
- $\ell_i \in \mathbb{R}_+$: Scheduled revenue track distance of trip i (km).
- $\rho_i \in \mathbb{Z}_+$: Forecasted peak passenger seat demand for trip i .

2.2 Fleet Unit Attributes ($\forall u \in U$)

- $c_u \in \mathbb{Z}_+$: Nominal passenger seating capacity of unit u .
- $h_u \in D$: Initial home depot of unit u .
- $a_u \in \mathbb{R}_+$: Earliest availability timestamp of unit u at depot h_u .
- $\mu_u^{\text{init}} \in \mathbb{R}_+$: Accumulated mileage since the last major maintenance overhaul (km).
- $M_{\text{type}(u)} \in \mathbb{R}_+$: Maximum allowable operating distance prior to required maintenance inspection.

2.3 Network and Cost Parameters

- $t(s_1, s_2) \in \mathbb{R}_+$: Deadhead travel runtime between stations $s_1, s_2 \in S$ ($t(s, s) = 0$).
- $\delta(s_1, s_2) \in \mathbb{R}_+$: Deadhead track distance between stations $s_1, s_2 \in S$ ($\delta(s, s) = 0$).
- W_{deadhead} : Weight penalty per empty deadhead kilometer (1.0).
- W_{unit} : Fixed operating cost per activated trainset unit (150.0).
- W_{maint} : Penalty cost per kilometer exceeding the maintenance threshold (5.0).
- W_{demand} : Penalty cost per uncovered passenger seat demand (2.0).
- W_{timing} : Penalty cost per minute of infeasible transition shortfall (1000.0).
- M_{big} : Sufficiently large positive scalar used in disjunctive constraints.

3 Decision Variables

- $x_{u,i} \in \{0, 1\}$: Binary variable equal to 1 if unit u is assigned to trip i , and 0 otherwise.
- $y_{u,i,j} \in \{0, 1\}$: Binary variable equal to 1 if unit u serves trip j immediately following trip i .
- $y_{u,i}^0 \in \{0, 1\}$: Binary variable equal to 1 if unit u departs from depot h_u to serve trip i as its initial trip.
- $y_{u,i}^* \in \{0, 1\}$: Binary variable equal to 1 if trip i is the final trip of unit u returning to depot.
- $z_u \in \{0, 1\}$: Binary variable equal to 1 if rolling stock unit u is activated.
- $s_i^{\text{dem}} \geq 0$: Continuous slack variable representing uncovered passenger demand on trip i .
- $v_{u,i,j}^{\text{time}} \geq 0$: Continuous slack representing timing shortfall on transition (i, j) for unit u .
- $m_{u,i} \geq 0$: Continuous state variable representing the cumulative mileage of unit u upon completing trip i .
- $v_u^{\text{maint}} \geq 0$: Continuous slack representing maintenance mileage limit violation for unit u .

4 Mathematical Formulation

4.1 Objective Function

The objective minimizes total operational costs, fleet deployment expenses, passenger demand shortfalls, and constraint penalties:

$$\begin{aligned} \min \quad \mathcal{J} = & W_{\text{deadhead}} \cdot \mathcal{D} + W_{\text{unit}} \cdot \mathcal{U} + W_{\text{maint}} \cdot \mathcal{M} \\ & + W_{\text{demand}} \cdot \mathcal{P} + W_{\text{timing}} \cdot \mathcal{V}_{\text{time}} \end{aligned} \quad (1)$$

where the individual cost components are defined as:

$$\mathcal{D} = \sum_{u \in U} \left[\sum_{i \in T} \delta(h_u, o_i) y_{u,i}^0 + \sum_{i \in T} \sum_{j \in T \setminus \{i\}} \delta(d_i, o_j) y_{u,i,j} \right] \quad (2)$$

$$\mathcal{U} = \sum_{u \in U} z_u \quad (3)$$

$$\mathcal{M} = \sum_{u \in U} v_u^{\text{maint}} \quad (4)$$

$$\mathcal{P} = \sum_{i \in T} s_i^{\text{dem}} \quad (5)$$

$$\mathcal{V}_{\text{time}} = \sum_{u \in U} \sum_{i \in T} \sum_{j \in T \setminus \{i\}} v_{u,i,j}^{\text{time}} \quad (6)$$

4.2 Constraints

1. Passenger Demand Satisfaction Each trip must satisfy passenger demand through assigned trainset capacities, with slack penalized in the objective:

$$\sum_{u \in U} c_u x_{u,i} + s_i^{\text{dem}} \geq \rho_i, \quad \forall i \in T \quad (7)$$

2. Flow Conservation along Unit Circulation Chains Every assigned unit must enter and exit each assigned trip exactly once:

$$y_{u,i}^0 + \sum_{j \in T \setminus \{i\}} y_{u,j,i} = x_{u,i}, \quad \forall u \in U, \forall i \in T \quad (8)$$

$$y_{u,i}^* + \sum_{j \in T \setminus \{i\}} y_{u,i,j} = x_{u,i}, \quad \forall u \in U, \forall i \in T \quad (9)$$

3. Depot Departure and Activation Linking A unit can depart from its depot at most once, and may only be assigned trips if activated:

$$\sum_{i \in T} y_{u,i}^0 \leq z_u, \quad \forall u \in U \quad (10)$$

$$x_{u,i} \leq z_u, \quad \forall u \in U, \forall i \in T \quad (11)$$

4. Temporal Feasibility and Turnaround Bounds For initial depot pull-out:

$$a_u + t(h_u, o_i) - \text{dep}_i \leq M_{\text{big}}(1 - y_{u,i}^0), \quad \forall u \in U, \forall i \in T \quad (12)$$

For consecutive trip transitions (i, j) :

$$\text{arr}_i + \tau_i + t(d_i, o_j) - \text{dep}_j \leq v_{u,i,j}^{\text{time}} + M_{\text{big}}(1 - y_{u,i,j}), \quad \forall u \in U, \forall i, j \in T, i \neq j \quad (13)$$

5. Maintenance Mileage Accumulation Recurrence Cumulative mileage accumulates along the circulation chain from the initial depot departure through each successive revenue and deadhead trip. The recurrence relations are formulated as follows:

For the initial trip departing from depot h_u :

$$m_{u,i} \geq \mu_u^{\text{init}} + \delta(h_u, o_i) + \ell_i - M_{\text{big}}(1 - y_{u,i}^0), \quad \forall u \in U, \forall i \in T \quad (14)$$

For subsequent consecutive transitions (i, j) :

$$m_{u,j} \geq m_{u,i} + \delta(d_i, o_j) + \ell_j - M_{\text{big}}(1 - y_{u,i,j}), \quad \forall u \in U, \forall i, j \in T, i \neq j \quad (15)$$

The maintenance violation slack v_u^{maint} captures any distance operated beyond the permissible limit:

$$v_u^{\text{maint}} \geq m_{u,i} - M_{\text{type}(u)} - M_{\text{big}}(1 - x_{u,i}), \quad \forall u \in U, \forall i \in T \quad (16)$$

$$v_u^{\text{maint}} \geq 0, \quad \forall u \in U \quad (17)$$

6. Integrality and Variable Domains

$$x_{u,i} \in \{0, 1\}, \quad \forall u \in U, \forall i \in T \quad (18)$$

$$y_{u,i,j} \in \{0, 1\}, \quad \forall u \in U, \forall i, j \in T \quad (19)$$

$$y_{u,i}^0, y_{u,i}^* \in \{0, 1\}, \quad \forall u \in U, \forall i \in T \quad (20)$$

$$z_u \in \{0, 1\}, \quad \forall u \in U \quad (21)$$

$$s_i^{\text{dem}} \geq 0, \quad \forall i \in T \quad (22)$$

$$v_{u,i,j}^{\text{time}} \geq 0, \quad \forall u \in U, \forall i, j \in T \quad (23)$$

$$m_{u,i} \geq 0, \quad \forall u \in U, \forall i \in T \quad (24)$$